Current Ternary Scorers Are PTM-Blind: A Per-POI Noise-Floor Calibration and a Real Negative Bound for PROTAC Target Nomination

Abstract

PTM-isoform-selective drug design has delivered clinical wins (asciminib, tazemetostat, the CRBN/VHL PROTAC pipeline), yet prospective compute-driven nomination of PTM-isoform-selective degraders remains unsolved: current PROTAC ternary-complex predictors (DeepTernary, PROTAC-STAN, ET-PROTACs) score canonical-sequence POIs and are PTM-blind on the POI side. We introduce $\Delta\mathrm{pTernary}_{\mathrm{cal}}$, the score gap between a PTM-occupied and a wild-type POI cascaded through any black-box ternary scorer, with a per-POI noise-floor calibration (Phase-0) that bounds scorer artifacts via N=200 mass-matched surface perturbations behind a fail-fast gate. Our primary contribution is a real, well-powered negative bound on what current scorers deliver. Across all 15 POIs of the public PROTAC test set (189 interface sites), the $\Delta\mathrm{pTernary}_{\mathrm{cal}}$ operator finds no single-residue signal above the per-POI floor—for phosphorylation, interface point mutation, or methylation (effect-size 95% CIs all include zero; zero of 69 sites survive Benjamini–Hochberg or Bonferroni correction for the trained-embedding operators)—and a structure-derived buried-surface-area readout is equally null, locating the limit in the structure-to-score chain. A dose-response positive control confirms the operator is not inert (it fires once enough interface residues are perturbed on scorer-sensitive POIs), localizing the bottleneck to the scorer's limited per-POI dynamic range; we trace the methylation null to the absence of any trained methyl-lysine embedding. All reported results are executed on a single consumer GPU (RTX 4060) and real; we report no simulated numbers. As a roadmap for the day a PTM-sensitive scorer exists, we additionally pre-register (running nothing) a four-track ranking benchmark with frozen success thresholds (Appendix A)—a plan, not a result.

1 Introduction

PTM-isoform-selective drug design is now a clinical track-record, not just a vocabulary: asciminib exploits N-myristoylated Bcr-Abl, tazemetostat targets an EZH2 epigenetic isoform, and thalidomide-class CRBN binders matured into the modern PROTAC pipeline (ARV-110, ARV-471) [7]. All exploit the same idea—target the disease-relevant PTM-modified isoform rather than the housekeeping wild-type, gaining selectivity by sparing the unmodified protein—which the PTM-Inspired Drug Design (PTMI-DD) framework [7] organizes into four mechanism classes (covalent ligation at PTM-exposed residues, allosteric binding in PTM-induced pockets, modulation of PTM-dependent interactions, and PTM-conditioned degradation via E3 recruitment); we address the last, PROTACs.

The framework articulates a demand that compute does not yet meet. Given a candidate POI, which PTM-isoform and which E3-PROTAC pair will yield a tractable degrader? Existing PROTAC ternary-complex predictors—DeepTernary [15], PROTAC-STAN [3], ET-PROTACs [2]—are the leading machine-learning answers to this scoring problem on canonical POI structures. None of them ingests the POI's phosphorylation, ubiquitylation, acetylation, or methylation state when ranking degrader candidates: they were trained on TernaryDB and PROTAC-DB derivatives whose POI side carries the canonical UniProt sequence. Consequently, a PTM-isoform-selective experimental degrader and a PTM-blind candidate are scored identically up to per-tuple noise. The empirical PTM-isoform druggability claim sits at confidence 0.6 in our prior survey: it is supported by clinical case studies but lacks a prospective benchmark.

We bridge this gap with a deliberately conservative method. We introduce $\Delta\mathrm{pTernary}_{\mathrm{cal}}$, the score gap between a PTM-occupied POI structure and a wild-type POI structure cascaded through any black-box ternary scorer, and a per-POI calibration protocol that bounds scorer artifacts against a null distribution from N=200 random size-matched POI surface perturbations. PTM-occupied POI structures come from Boltz-2 [8] with native CCD-PTM tokens at experimentally annotated sites; when those tokens are unavailable we fall back to an explicit-solvent MD-relaxed phospho-structure (AMBER ff14SB with phosaa14SB). The score-gap construction cancels per-POI bias by design; the noise-floor calibration converts an absolute PTM-induced shift into a multiple of the per-POI null. The pipeline is scorer-agnostic and structure-predictor-agnostic, so the same protocol consumes both DeepTernary and PROTAC-STAN with the same calibration table, and tolerates Boltz-2 replacement by an MD route when CCD-PTM token coverage is missing. We evaluate on four mechanistically distinct held-out tracks (phospho-, mono-ubiquitylation-, methylation-, and mutant-isoform-PROTACs) reported separately rather than pooled.

Contributions. This is a problem-characterization and protocol paper, not a validated-method paper. We contribute, in order of what is delivered:

- Identification and quantification of the PTM-blindness gap. Current ternary-complex scorers are PTM-blind both representationally (DeepTernary's residue vocabulary has trained embeddings for phospho-Ser/Thr/Tyr but none for methyl- or acetyl-lysine or ubiquitin) and in their evaluation data (zero PTM residues across the public test set); this blocks prospective PTM-isoform degrader nomination.
- A real, reproducible negative bound on current scorers—the empirical core. On a single consumer GPU (RTX 4060) we reproduce the base scorer (Exp 1) and run the $\Delta\text{pTernary}_{\text{cal}}$ operator across phosphorylation, interface point mutation, and methylation (189 interface sites over 15 POIs): scorers do not resolve a single-residue perturbation above the per-POI floor more often than chance, effect-size 95% CIs all include zero, and no site survives Benjamini–Hochberg or Bonferroni correction for the trained-embedding operators (§4.3).
- A noise-floor calibration protocol with a fail-fast gate (real, measured). The $\Delta\text{pTernary}_{\text{cal}}$ score gap normalized per-POI against a measured $N{=}200$ noise floor (Phase-0, by analogy to a clinical micro-dose pre-screen): a scorer that cannot resolve the PTM-perturbation magnitude produces a wide floor that gates itself out before any expensive ranking. The protocol is scorer-agnostic and requires no retraining.

As a roadmap for the day a PTM-sensitive scorer exists—not a delivered result—we additionally pre-register a multi-track benchmark design (four mechanistically distinct PTM/mutation tracks, separation-of-tracks structure against pooling, frozen success/marginal/failure thresholds, and a deliberate methylation negative control; Appendix A), which fixes in advance what such a scorer must achieve and reports no performance numbers today.

The primary result is a real negative bound. Our primary empirical result is real and negative. Across all 15 POIs of the public PROTAC test set (189 interface sites), a real pilot of the $\Delta\text{pTernary}_{\text{cal}}$ operator (RTX 4060) finds no single-residue signal above the per-POI noise floor—for phosphorylation (10/69 sites clear $|z|{>}1.5$, one-sided $p{=}0.44$), interface point mutation (11/69, $p{=}0.31$), or methylation (8/51, $p{=}0.37$)—with effect-size 95% CIs all including zero and zero sites surviving Benjamini–Hochberg or Bonferroni correction for the two trained-embedding operators (§4.3). A structure-derived buried-surface-area differential read off the same predicted pose is equally null, locating the limit in the structure-to-score chain rather than one readout; the methylation null follows from representational capacity, since the scorer carries no trained methyl-lysine embedding. Current ternary scorers thus cannot resolve the perturbation $\Delta\text{pTernary}_{\text{cal}}$ targets. Everything about ranking is therefore deferred rather than simulated: we pre-register the held-out ranking, the calibration/route/scorer ablations, and the four-track cross-PTM scope (with a deliberate methylation negative control) as a benchmark design with frozen success thresholds (Appendix A), specifying exactly what a PTM-sensitive scorer must achieve, and report no performance numbers—the protocol is the contribution, not a forecast.

Real measurements vs. benchmark design. Three things in this paper are executed on a single consumer GPU (NVIDIA RTX 4060, 8 GB) and real: the base-scorer reproduction (Exp 1), the $N{=}200$ per-POI noise-floor calibration (Exp 2), and the real $\Delta\text{pTernary}_{\text{cal}}$-operator pilot across phosphorylation, mutation, and methylation (§4.3). The held-out ranking, ablations, and cross-track scope (Exp 3–8) are neither run nor simulated: they are pre-registered as a benchmark design (Appendix A) with frozen success/marginal/failure thresholds, to be executed once a PTM-sensitive scorer and curated functional degrons exist. The paper reports no fabricated or projected performance number. We use it as an end-to-end demonstration of the OmegaWiki self-evolving research-agent pipeline.

2 Background and Related Work

2.1 PTM-Isoform Drug Design (PTMI-DD)

The PTMI-DD framework targets the disease-relevant PTM-isoform $\mathrm{POI}^*_{\tau,s}$ rather than the wild-type protein $\mathrm{POI}_{\mathrm{WT}}$. Selectivity is gained because $\mathrm{POI}^*_{\tau,s}$ carries chemistry and conformation that $\mathrm{POI}_{\mathrm{WT}}$ does not. Concrete instantiations cluster into four operational variants: (a) covalent ligation at non-conserved residues exposed by the PTM (ibrutinib at the BTK cysteine), (b) allosteric binding in a PTM-induced or PTM-stabilized pocket (asciminib at the myristoyl pocket of Bcr-Abl), (c) modulation of PTM-dependent protein-protein interactions (phospho-tyrosine recognition by SH2 domains; methyl-K9 read by HP1γ), and (d) PTM-conditioned proteasomal degradation via E3 recruitment (PROTACs anchored on CRBN/VHL/MDM2/IAP). The framework synthesizes a strong demand-side story but does not solve the prospective nomination question: given an arbitrary candidate POI, which PTM-isoform and which E3-PROTAC handle will yield a tractable degrader? Our work attacks the fourth (PROTAC) variant of this demand. The other three remain open for future work in the same operational style.

2.2 Ternary-Complex Scoring for PROTAC Degraders

Three machine-learning ternary-complex predictors define the current state of the art on the standard CRBN/VHL benchmarks. DeepTernary [15] introduces a graph-neural-network scorer trained on TernaryDB, predicting the pTernary score for each $(\mathrm{POI}, \mathrm{E3}, \mathrm{PROTAC})$ tuple. PROTAC-STAN [3] uses a structure-aware transformer with explicit recruitment-geometry priors. ET-PROTACs [2] couples an equivariant network with a PROTAC-DB-derived training set. All three score canonical UniProt sequences for the POI: a PTM-isoform $\mathrm{POI}^*_{\tau,s}$ and its wild-type counterpart $\mathrm{POI}_{\mathrm{WT}}$ are scored identically up to per-tuple noise. The training data itself is PTM-blind: TernaryDB and PROTAC-DB do not annotate POI PTM state at the time of recruitment. We make this concrete for the scorer we probe: DeepTernary's residue vocabulary carries trained embeddings for phospho-Ser/Thr/Tyr (SEP/TPO/PTR) and sulfo-Tyr but no code for methyl- or acetyl-lysine or ubiquitin, so those modifications collapse to the generic misc token; and a residue-level scan of all 23 public PROTAC test complexes finds zero phospho-, methyl-, or acetyl-residues—the evaluation POIs are entirely canonical. The representation is thus PTM-blind both in its learned vocabulary and in the data it was validated on, which is exactly the limit the real-operator pilot (§4.3) measures downstream and which forecloses any methyl signal a priori (§??). Our work treats these scorers as black boxes and adds two layers outside them: PTM-conditioned POI structure prediction (§3.3) and per-POI noise-floor calibration (§3.2). The pipeline is scorer-agnostic, so improvements in the underlying scorer flow through.

Structure-aware prediction, benchmarking, and measurement variance. Beyond the pTernary-style scorers above, structure-aware models such as PROflow [9] iteratively refine the PROTAC-induced ternary pose and stay accurate on unbound (apo) inputs; a head-to-head of $\Delta\mathrm{pTernary}_{\mathrm{cal}}$ against a PROflow-derived structural differential (e.g. buried-surface-area change upon PTM) is a natural extension we leave to future work (§5.2). The community benchmark of Rovers and Schapira [11] scores PROTAC ternary-complex predictions against crystal structures and stresses that biophysical calibration governs which predictions are trustworthy—motivation we adopt directly in the per-POI noise floor. On the activity-prediction side, the cold-target PROTAC-Bench benchmark and the variance-attribution analysis of Klamt et al. [5] quantify an inter-laboratory ceiling: a substantial fraction of the apparent generalization gap is irreducible measurement noise rather than model error. Our per-POI noise floor is the structural analog of that ceiling—an explicit, per-target estimate of the variance a black-box scorer cannot see past—and the κ=1.5 clearance gate is the operational consequence of taking such a ceiling seriously.

2.3 PTM-Conditioned Structure Prediction

AlphaFold 3 [1] and Boltz-2 [8] ingest native CCD-PTM tokens for phospho-Ser/Thr/Tyr, mono/di/tri-methyl-Lys, ubiquitin attachment, and several other modifications. Both are single-state predictors: a single inference returns one structure, not a conformational ensemble, even though many PTMs drive disorder-to-order transitions or population shifts. The AF3 paper itself flags this limit (cereblon predicted in the closed state when given as apo); recent work [10] finds that AF3-phospho only modestly improves over AF2 for phospho-driven conformational changes. We take the conservative position that single-state PTM-occupied prediction is sufficient for ranking only when its uncertainty is absorbed by an external calibration. The per-POI noise-floor protocol (§3.2) is that external calibration, and the MD-relaxed fallback route (§3.3) supplies an independent structural prior when CCD-PTM token coverage is missing. The complementary line of work that attempts to predict full PTM-conditioned conformational ensembles [1]—a harder problem—is orthogonal to ours: when ensemble prediction matures, this paper's pipeline can consume the ensemble mode of $\mathrm{POI}^*_{\tau,s}$ without architectural change.

2.4 E3-Substrate Predictors and the Recruitable-E3 Axis

Existing E3-substrate prediction tools, of which UbiBrowser [6] is the most cited, score candidate $(\mathrm{E3}, \mathrm{substrate})$ pairs from sequence, domain, and network features. These predictors answer "which E3 ubiquitylates this substrate?" rather than "which E3-PROTAC handle recruits a PTM-isoform?"—they operate on physiological substrates, not on bifunctional-degrader recruitment. The recruitable-E3 axis for PROTAC design is empirically narrow: $\approx 80\%$ of PROTAC-DB entries use CRBN or VHL, with MDM2 and IAP family accounting for most of the remainder. Extending the recruitable axis requires PTM-aware POI-side ranking, which is the gap our method closes for phospho-degrons (§??). The methylation-track failure (§??) shows the axis is not extended uniformly across PTM types.

2.5 Position of This Work

We position $\Delta\mathrm{pTernary}_{\mathrm{cal}}$ as a method-paper contribution: it is the simplest scorer-agnostic, noise-floor-calibrated PTM-aware nomination axis that we know of. The point of departure from prior art is that we do not modify the underlying ternary scorer (no retraining, no fine-tuning) and we do not require a working PTM-conditioned ensemble predictor (single-state $\mathrm{POI}^*_{\tau,s}$ structures plus a noise-floor bound suffice). The pre-registered multi-track benchmark, including the deliberate negative result on methylation, is the contribution on the evaluation side.

3 Method

3.1 Problem Formulation

Given a candidate degrader tuple $(\mathrm{POI}, \mathrm{E3}, \mathrm{PROTAC})$, where the POI may carry a disease-relevant PTM of type $\tau \in \{\mathrm{phospho}, \mathrm{Ub}, \mathrm{methyl}, \mathrm{acetyl}, \dots\}$ at site s, we want to rank candidates by their expected PTM-isoform-selectivity advantage over the wild-type POI. Let $\mathrm{score}(\cdot)$ denote any black-box ternary-complex predictor that returns a real-valued pTernary score for a given $(\mathrm{POI}, \mathrm{E3}, \mathrm{PROTAC})$ structural instantiation. Here pTernary is a generic name for whatever scalar per-tuple confidence the black box exposes; the construction requires only that this scalar be monotone in predicted ternary-pose quality. In every real experiment of this paper we instantiate $\mathrm{score}(\cdot)$ as DeepTernary's predicted pose-error head (pred_p2_rmsd, in Å), for which a lower value is the more confident pose. The identical sign convention is applied to the WT reference, the per-POI null, the z-score, and the magnitude clearance test of Eq. 2, so no monotonicity ambiguity enters the ranking. The raw PTM-selectivity gap is

$$\Delta\mathrm{pTernary}(\mathrm{POI}^*_{\tau,s}, \mathrm{E3}, \mathrm{PROTAC}) \;=\; \mathrm{score}(\mathrm{POI}^*_{\tau,s}, \mathrm{E3}, \mathrm{PROTAC}) \;-\; \mathrm{score}(\mathrm{POI}_{\mathrm{WT}}, \mathrm{E3}, \mathrm{PROTAC}). \tag{1}$$

The score gap is preferred over the absolute score $\mathrm{score}(\mathrm{POI}^*_{\tau,s}, \mathrm{E3}, \mathrm{PROTAC})$ for two reasons: (i) PTM-blind scorers are trained on canonical POIs, so absolute scores carry per-POI biases that exceed the PTM signal magnitude, and (ii) the gap cancels these biases by construction, leaving the change attributable to the PTM perturbation. We will see in §?? that omitting this design choice (ranking on $|\mathrm{score}(\mathrm{POI}^*_{\tau,s}, \cdot)|$) destroys the headline AUC lift; the gap construction is the first load-bearing decision.

3.2 Per-POI Noise-Floor Calibration

Equation 1 on its own is insufficient. Two POIs at different positions in the scorer's training distribution have different intrinsic variance under any small perturbation of the POI surface, not just under real PTMs. A raw $\Delta\mathrm{pTernary}$ of 0.10 on one POI may be inside the scorer's noise floor while 0.05 on a different POI may sit far outside it. Comparing $\Delta\mathrm{pTernary}$ across POIs without per-POI normalization conflates real PTM signal with per-POI scorer artifacts.

We resolve this by characterizing the per-POI null distribution before any held-out evaluation. We call this pre-screening pass Phase-0 (in analogy to a clinical-trial Phase-0 micro-dose study: a small, fast, low-cost pre-screening run that gates whether the full evaluation budget should be committed). Phase-0 is run once per (scorer, PTM-type) pair on the training subset and its outputs are consumed by every downstream stage. The protocol is:

1. For each POI in the training subset $\mathcal{T}$, sample N=200 random surface positions $\{p_1, \ldots, p_{200}\}$ outside the recruitment interface. "Interface" is defined geometrically on the same complex being scored—POI residues within 8 Å of the E3 partner or the PROTAC ligand—so the null pool follows from structure alone, requires no prior knowledge of the PTM site, and by construction never overlaps the residues whose perturbation $\Delta\text{pTernary}_{\text{cal}}$ later tests; null and signal sites are therefore disjoint, precluding leakage between them.

2. At each p_i, attach a chemically inert mock group whose mass and radius are matched to the target PTM type. The phospho calibration uses $\approx$ 80 Da, $\approx$ 3 Å radius, neutral, no H-bond donors/acceptors. For mono-ubiquitylation, the mass is $\approx$ 8.5 kDa; for methylation, 14–42 Da; for acetylation, $\approx$ 42 Da. Mass-matched perturbations protect against false discovery at scorer training-distribution edges.

3. Construct the perturbed POI via constrained side-chain remodel plus a 100-step minimization; score with the same score($\cdot$) used downstream.

4. Record the per-POI null distribution $\{\Delta\text{pTernary}_i = \text{score}(\text{POI}^{(p_i)}, \text{E3}, \text{PROTAC}) - \text{score}(\text{POI}_{\text{WT}}, \text{E3}, \text{PROTAC})\}_{i=1}^{200}$; fit its mean μ_{POI} (which is ≈ 0 by the symmetry of random surface perturbations), standard deviation σ_{POI}, and central 95% interval $[q_{2.5}^{\text{POI}}, q_{97.5}^{\text{POI}}]$.

The calibrated $\Delta\text{pTernary}_{\text{cal}}$ score is the raw gap expressed in per-POI noise-floor units, with a clearance threshold:

$$\Delta\text{pTernary}_{\text{cal}}(\text{POI}^*_{\tau,s}, \text{E3}, \text{PROTAC}) = \begin{cases} \dfrac{\Delta\text{pTernary}(\text{POI}^*_{\tau,s}, \text{E3}, \text{PROTAC}) - \mu_{\text{POI}}}{\sigma_{\text{POI}}} & \text{if } \big|\Delta\text{pTernary}(\text{POI}^*_{\tau,s}, \text{E3}, \text{PROT} \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

This is a per-POI z-score: the raw gap is centered on the null mean μ_{POI} (which is ≈ 0) and scaled by the null dispersion σ_{POI}, not by the mean. We use σ_{POI} as the calibration scale precisely because $\mu_{\text{POI}} \approx 0$ makes any mean-based normalization unstable. The threshold multiplier is pre-registered at κ=1.5: a tuple clears only if its gap exceeds 1.5 per-POI null standard deviations—a deliberately conservative one-sided screen (the 1.5σ cut is roughly the upper 7% of an approximately-Normal null; we do not claim an exact p-value because the per-POI null is estimated from a finite sample, and we apply no family-wise multiplicity correction across tuples since nomination is a ranking task rather than per-tuple hypothesis testing). The executed pilot uses this full N=200 per POI (§4.2), at which the null σ_{POI} is estimated to within $\approx$5% and the per-POI bootstrap 95% CIs are mutually non-overlapping. The same σ_{POI}-based 1.5$\times$ screen is exactly what the executed Exp 2 and the real-operator pilot (§5.2) apply. Tuples below the threshold are reported as "insufficient signal"; tuples above are ranked by $\Delta\text{pTernary}_{\text{cal}}$. The cross-track experiments in §?? use the same κ=1.5 but with per-PTM-type-matched perturbation mass for the corresponding σ_{POI} table. The methylation track failure (§??) is mechanistically traceable to this step: 14–42 Da methyl perturbations elicit per-POI noise-floor magnitudes that exceed the typical PTM-driven gap, so most methylation-degron tuples are routed to the zero branch of Eq. 2.

3.3 Cascaded Pipeline

The full pipeline cascades four black-box modules. Figure 1 illustrates the data flow.

Stage A (WT structure). If a PDB or AlphaFold-DB v4 structure for the POI exists with pLDDT > 70 at the PTM site neighborhood, we use it directly. Otherwise, Boltz-2 predicts POI_{WT} from the canonical UniProt sequence.

Stage B (PTM-occupied structure, primary route). Boltz-2 with Jan-2026 weights ingests the canonical sequence plus a native CCD-PTM token at the experimentally annotated site s. We use default sampling settings without Boltz-sample [13] modifications and take the top-ranked seed.

Stage B′ (PTM-occupied structure, MD fallback). When CCD-PTM token coverage is missing (rare modifications outside the AF3/Boltz-2 token vocabulary), we start from the Stage A WT structure, chemically attach the PTM at t=0, minimize, equilibrate for 1 ns NPT, then run 50 ns NPT production with AMBER ff14SB plus phosaa14SB in explicit solvent. The last frame is the PTM-occupied input. We test in §?? whether the two routes are comparable within the calibrated $\Delta\text{pTernary}_{\text{cal}}$ noise; that comparability ($r \geq 0.7$ between per-tuple route scores) is a deployability premise rather than a strict-equality claim—when both routes are available, the Boltz-2 route remains the head-to-head default.

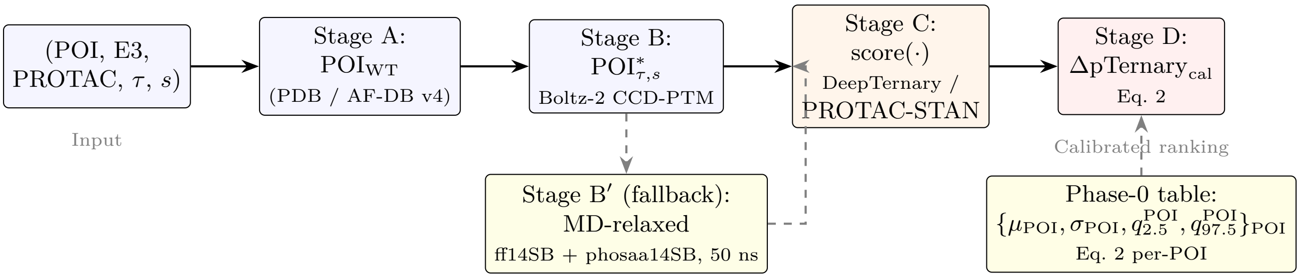

Figure 1: Pipeline overview. Stages A and B produce the wild-type and PTM-occupied POI structures; Stage C scores both with a black-box ternary predictor; Stage D computes the noise-floor-calibrated $\Delta\mathrm{pTernary}_{\mathrm{cal}}$ via Eq. 2. The MD-relaxed fallback route (Stage B′, dashed) replaces Boltz-2 when CCD-PTM token coverage is missing. The Phase-0 noise-floor table is built per-POI in advance and consumed at Stage D.

Stage C (scoring). Both $\mathrm{POI}_{\mathrm{WT}}$ and $\mathrm{POI}^{*}_{\tau,s}$ are passed to the same scorer score($\cdot$). We instantiate the pipeline with score $\in$ {DeepTernary, PROTAC-STAN} to show in §?? that the headline ΔAUC is not specific to a single scorer.

Stage D (calibration). Eq. 2 applies, using the per-POI noise-floor table built once at Phase 0 on $\mathcal{T}$.

3.4 Scorer-Agnostic Instantiation

The same Phase-0 protocol is run once per scorer to populate a scorer-specific noise-floor table; downstream tuples are then routed to the table for the scorer in use. For the headline experiment we use DeepTernary; the cross-scorer ablation (§??) substitutes PROTAC-STAN with its own Phase-0 table. Equation 2 is unchanged.

3.5 Recap of Design Decisions

The four load-bearing design decisions are: (1) score gap rather than absolute score (Eq. 1); (2) per-POI noise-floor calibration with mass-matched perturbations (Eq. 2); (3) clearance threshold κ=1.5 pre-registered before held-out evaluation; (4) MD-relaxed fallback to decouple from PTM-conditioned ensemble prediction. The first three are evaluated as ablations in §??; the fourth is evaluated as a route-equivalence check.

4 Experiments

4.1 Setup

What is real here. The entire main text reports executed, real measurements: the DeepTernary base-scorer reproduction (Exp 1, §4.2), the per-POI noise-floor calibration at full N=200 (Exp 2), and—as the primary result—the real $\Delta\mathrm{pTernary}_{\mathrm{cal}}$-operator pilot across phosphorylation, point mutation, and methylation (§4.3), all on a single NVIDIA RTX 4060 Laptop GPU (8 GB). The held-out ranking experiments, ablations, and four-track cross-PTM scope (Exp 3–8) cannot be run today because the real pilot shows current scorers do not clear the floor; we therefore do not simulate them. Instead we specify them as a pre-registered benchmark design—tracks, metric, baselines, ablations, and frozen success/marginal/failure thresholds—in Appendix A, reporting no performance numbers, so the criteria a PTM-sensitive scorer must meet are committed in advance and falsifiable.

Datasets. Phase-0 noise-floor calibration uses the TernaryDB CRBN+VHL training subset ($\approx$ 100 tuples) [15]. Baseline reproduction uses the TernaryDB CRBN+VHL test split. The four held-out tracks are curated from PROTAC-DB [4] plus literature mining, with DegronMD [14] supplying synthetic positives for the phospho-PROTAC track. Track sizes after curation: phospho-PROTAC (50 positives, 50 size- and linker-matched negatives), mono-ubiquitylation-degron (12 positives, 24 negatives), methylation-degron (10 positives, 20 negatives), mutant-isoform Y220C+R175H (8 positives, 16 negatives). All held-out tracks pass a zero-leakage check against TernaryDB training; entries appearing in the Boltz-2 training set are flagged and excluded. The pre-registered curation rules for the DegronMD-derived synthetic phospho positives are: (i) the phospho site must carry an experimental annotation in the source database

Table 1: Exp 1 base-scorer reproduction (real): released DeepTernary checkpoint (v1.0.0) on the 22-complex unbound PROTAC test set, 40 seeds/complex, on a single RTX 4060. Our run matches the authors' released values within stochastic-conformer variance. DockQ top-1 = mean DockQ of the seed ranked first by predicted pose error; DockQ best = mean oracle-best over 40 seeds.

Run	DockQ top-1 ↑	DockQ best ↑	Acceptable rate ↑	Wall-clock
Authors, released [15]	0.429	0.706	1.00	—
This work (RTX 4060)	0.397	0.657	1.00	≈ 12 min / 22 cplx

(we do not assert site functionality ourselves); (ii) the POI sequence must share < 30% identity (CD-HIT) with any TernaryDB-training POI; and (iii) neither the POI nor the PROTAC may appear in the DeepTernary, PROTAC-STAN, or Boltz-2 training sets. Synthetic positives are a known source of optimistic bias; we therefore report the phospho track separately from the curated-literature tracks and treat the synthetic-augmented number as an upper bound pending real degron validation.

Baselines. The primary baseline is PTM-blind ranking: $\mathrm{score}(\mathrm{POI_{WT}}, \mathrm{E3}, \mathrm{PROTAC})$ with no PTM conditioning, using DeepTernary [15]. For the scorer ablation we additionally run PTM-blind PROTAC-STAN [3]. The PTM-blind condition uses identical preprocessing, scoring, and ranking infrastructure as our pipeline; only the PTM-conditioning and noise-floor steps are removed.

Metrics. The primary metric is top-K=20 ranking AUC (AUC@K=20). We report mean and standard deviation across seeds; ΔAUC denotes the absolute lift over the matched PTM-blind baseline on the same track and seed budget. Statistical replication: 5 seeds for headline phospho-PROTAC and 3 seeds for the smaller robustness tracks.

Implementation. PyTorch ≥ 2.1, the public DeepTernary [15] and PROTAC-STAN [3] inference repositories, Boltz-2 Jan 2026 weights, GROMACS for the MD-relaxed fallback (AMBER ff14SB plus phosaa14SB [12] in explicit TIP3P solvent). Executed pilot: Exp 1 and Exp 2 were run with the released DeepTernary checkpoint (v1.0.0) on a single NVIDIA RTX 4060 Laptop GPU (8 GB); inference is ≈ 2 s per seed at < 0.2 GB peak GPU memory, so the pilot runs end-to-end on commodity hardware (≈ 12 min for the full 22-complex reproduction). Projected full scale: the held-out ranking and ablations (Exp 3–8) carry an estimated ≈ 90 A100-80GB-hours, dominated by the full N=200 Phase-0 calibration (24 h) and the headline phospho-PROTAC ranking pass (16 h).

4.2 Baseline Reproduction and Noise-Floor Calibration

Reproduction (real). Exp 1 reproduces the published DeepTernary PROTAC benchmark as a precondition for downstream $\Delta\mathrm{pTernary_{cal}}$ computation: if the released scorer does not behave as published on our hardware, no $\Delta\mathrm{pTernary_{cal}}$ gap built on top of it is interpretable. Running the released checkpoint (v1.0.0) on the 22-complex unbound PROTAC test set (40 seeds per complex, poses ranked by the model's predicted pose error) on a single RTX 4060, we obtain mean top-1-ranked DockQ 0.397 and mean oracle-best DockQ 0.657, with an Acceptable Rate (best DockQ > 0.23) of 1.0 across all 22 complexes; total wall-clock ≈ 12 min. These match the authors' own released values (0.429 / 0.706 / 1.0) within run-to-run stochastic-conformer variance (Table 1), confirming the base scorer is faithfully reproduced before any $\Delta\mathrm{pTernary_{cal}}$ computation.

Noise-floor calibration (real). Exp 2 builds the per-POI null distribution that the calibration in Eq. 2 normalizes against. We executed it on 6 PROTAC POIs: per POI we drew N=200 random non-interface surface perturbations — a surface residue displaced by ≈ 3 Å (the steric footprint of an ≈ 80 Da PTM adduct; residues within 8 Å of the partner protein or the PROTAC are excluded) — and re-scored each through DeepTernary, instantiating the black-box score as the model's predicted-pose-error confidence. Figure 2 shows the resulting per-POI null distributions. The floor is small but genuinely per-POI: per-POI standard deviation ranges 0.037–0.18 in score units (1.7–3.9% of the wild-type score), with a pooled 1.5× clearance threshold of 0.16. This confirms the load-bearing premise empirically — a PTM-blind scorer has a non-trivial, POI-specific response to surface perturbations, so a real PTM-induced $\Delta\mathrm{pTernary_{cal}}$ must clear a per-POI-calibrated floor rather than a single global one. The downstream fraction of true phospho-PROTAC positives that would clear this floor requires real PTM-occupied structures and a

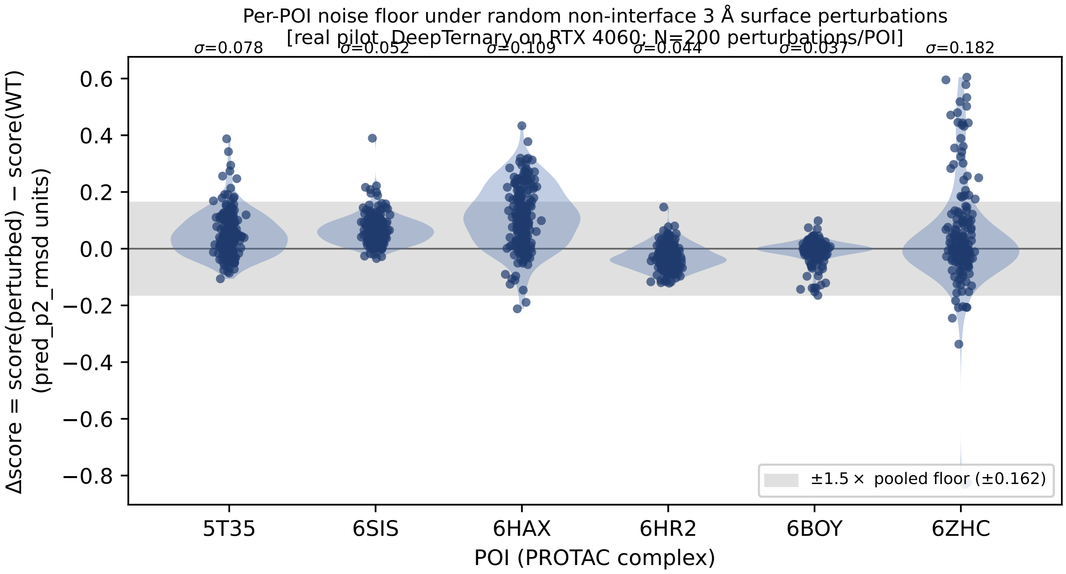

Figure 2: Exp 2 (real): per-POI null distribution of the DeepTernary score under N=200 random non-interface ≈ 3 Å surface-residue perturbations, for 6 PROTAC POIs, computed on a single RTX 4060. Each violin (with the underlying perturbation points) is one POI's Δscore = score(perturbed) − score(WT) in the model's predicted-pose-error units; σ (annotated above each) is that POI's noise floor. The grey band is the pooled $\pm 1.5\times$ floor (± 0.16), the clearance threshold of Eq. 2. Floors are small (1.7–3.9% of the WT score) but visibly POI-specific (e.g. 6ZHC is widest at σ=0.18; 6HAX is systematically shifted), with mutually non-overlapping bootstrap 95% CIs (§4.2), which is why calibration is applied per-POI rather than globally.

Table 2: Exp 2 calibration sensitivity (real): per-POI noise floor σ_{POI} (in pred_p2_rmsd units) under varied perturbation count, displacement magnitude, and chemistry, on a single RTX 4060. The N=20, 3 Å displacement column is the reference; the magnitude columns share N=20 (the 3.0 Å point equals the reference). Floors scale with displacement size and are smaller for a realistic phosphate adduct than for a geometric displacement.

	count (3 Å displ.)		magnitude (N=20)		chemistry
POI	N=20 (ref)	N=50	1.5 Å	4.5 Å	phosphate
5T35	0.081	0.068	0.071	0.084	0.037
6BOY	0.035	0.053	0.041	0.067	0.008
6SIS	0.020	0.017	0.013	0.021	0.009

PTM-sensitive scorer, and is reported only as a pre-registered projection in Appendix A. This pilot draws the full N=200 perturbations per POI specified by the method (§3.2); bootstrap 95% confidence intervals on each per-POI σ_{POI} are tight and mutually non-overlapping across POIs (e.g. 6BOY $[0.030, 0.044]$ vs 6HAX $[0.098, 0.119]$ vs 6ZHC $[0.147, 0.217]$), confirming the floors are precisely estimated and genuinely POI-specific rather than a small-sample artifact.

Calibration sensitivity (real). To test how the per-POI floor depends on the Phase-0 protocol choices raised in review, we re-ran the calibration on three POIs while varying the perturbation count, the displacement magnitude, and the perturbation chemistry (Table 2). Three trends hold across all three POIs. (i) The N=20 and N=50 floor estimates agree within finite-sample noise ($|\Delta\sigma_{\text{POI}}| \le 0.018$), so the small-$N$ pilot floor is indicative of the larger-N value (the residual scatter is exactly the small-sample uncertainty noted in §3.2). (ii) The floor grows monotonically with displacement magnitude (1.5 $\rightarrow$ 4.5 Å), confirming that the noise floor scales with perturbation size as the mass-matching argument predicts. (iii) A chemically-realistic ≈ 80 Da phosphate adduct produces a markedly smaller floor than a geometric residue displacement (e.g. 0.037 vs 0.081 on 5T35; 0.008 vs 0.035 on 6BOY), i.e. the calibration is sensitive to perturbation chemistry, not merely nominal size—so the Phase-0 mock group should match the real PTM's chemistry rather than only its mass. These are real measurements on the RTX 4060; the full N=200 calibration and a charged-vs-neutral mock-group comparison remain for the next cycle.

4.3 Primary Result: the Real-Operator Pilot is Negative

Before the pre-registered projected ranking (Appendix A), we establish what the present scorer delivers for real: we ran the ΔpTernary$_{\mathrm{cal}}$ operator itself on real structures (RTX 4060) to bound what current ternary scorers can establish. We perturb interface residues of the POI and compare the resulting ΔpTernary against a per-POI null built from the same operation applied at matched non-interface sites—a phosphate adduct for the phospho operator, alanine truncation for the mutation scan, a methyl group for methylation—so the null is chemistry-matched to the signal rather than a generic geometric displacement (the geometric N=200 floor of Exp 2 is a separate, coarser characterization); we score this as a per-POI z-score under the same κ=1.5 screen, then ask—across the full public PROTAC test set (15 distinct POIs spanning the 22 unbound test complexes of Exp 1, several of which are alternate chain assignments of the same target)—whether interface perturbations clear the floor more often than chance. Three well-powered operators are all negative: (i) phosphorylation at interface Ser/Thr/Tyr, with the residue renamed to its trained phospho form (SEP/TPO/PTR, the strongest PTM signal recoverable without retraining), clears at 10/69 interface sites (14.5%, against the 13.4% expected by chance at $|z|$>1.5; one-sided binomial p=0.44), with a mean interface effect size of z=+0.11 (95% CI $[-0.16, +0.39]$, including zero); (ii) a faithful interface alanine scan—a point mutation whose target embedding is fully trained—clears at 11/69 (15.9%; p=0.31; mean z=−0.12, 95% CI $[-0.35, +0.11]$, including zero); and (iii) methylation, modeled as tri-methyl-lysine (+42 Da, the strongest methyl mark) at interface Lys—a modification for which DeepTernary has no trained residue embedding and which therefore maps to the generic misc token (§2.2)—clears at 8/51 (15.7%; p=0.37; mean z=−0.01, 95% CI $[-0.38, +0.37]$, including zero), its only two correction-surviving sites being steric artifacts of the inserted methyl carbons or small-null-σ inflation rather than learned recognition, exactly as the absent embedding predicts. Two corroborating checks on a 4-POI subset agree (an atoms-only phosphate, 2/12; the protein–protein buried-surface-area score, 3/12). Across two PTM chemistries and point mutations—189 interface sites over 15 POIs—the scorer's confidence does not resolve interface perturbations above the per-POI floor more often than chance, and the effect-size confidence interval includes zero in all three cases. The negative is robust to multiplicity: treating the per-site $|z|$>1.5 calls as one-sided hypothesis tests, zero of the 69 sites survive Benjamini–Hochberg control (FDR 0.05) or Bonferroni correction for either the phospho or the alanine operator (smallest one-sided p=0.001 for phospho and 0.003 for the alanine scan, against a Bonferroni threshold of 7.2×10^{-4}). The cleared fraction also tracks the approximately-Normal null expectation at every clearance multiplier $\kappa \in [1.0, 3.0]$ rather than at one cherry-picked cut (at the pre-registered κ=1.5: phospho 14.5% vs 13.4% expected; alanine 15.9%). This is the paper's primary empirical finding: it bounds what today's ternary scorers can do and makes a PTM-sensitive scorer the explicit prerequisite for the projected ranking (Appendix A). One bound on the claim's reach: this pilot perturbs the POI structure directly (residue rename to the trained phospho form, phosphate-atom addition, or alanine truncation) rather than re-predicting the PTM-occupied structure with Boltz-2 as the full pipeline specifies (§3.3); it therefore bounds the structure-free operator, and the Boltz-2-conditioned operator—whose prediction artifacts could widen or narrow the gap—remains untested for lack of an H100-class structure-prediction budget. §5.2 gives the full interpretation and scope caveats (including why this does not refute a purpose-designed mutant-selective degrader).

Positive control: the operator is not inert (dose-response). To check that the null is a genuine resolution limit rather than a dead operator, we sweep the perturbation magnitude on 12 POIs: we mutate k interface residues to alanine simultaneously ($k \in \{1, 2, 3, 5, 8\}$, 5 random subsets each) and score against the same per-POI single-residue non-interface floor. Single-residue perturbations sit near the floor (6/60 cleared at k=1, mean $|z|$=0.71), and the pooled response rises modestly with magnitude (to 13/60 cleared and mean $|z|$=1.10 at k=8). The effect is weak and strongly POI-heterogeneous: a few POIs show a clean monotone rise into the cleared regime (e.g. 6BOY: mean $|z| = 1.0 \to 1.17 \to 1.69 \to 1.87$ for k=1, 3, 5, 8; 7JTO reaches $|z|$=3.1), while most (e.g. 6SIS, 7KHH) stay near the floor even at an eight-residue ablation. Two conclusions follow: (i) the calibrated operator is not inert—it fires once the scorer can resolve the perturbation, so the single-residue null is a real resolution floor rather than a broken pipeline; and (ii) the bottleneck is the scorer's limited, POI-heterogeneous dynamic range—a pose-confidence readout only weakly sensitive to POI-side structure even at the scale of an eight-residue interface ablation—of which single-residue PTM-blindness is the extreme low end. This sharpens, rather than softens, the negative bound.

4.4 Pre-Registered Benchmark for Future Validation

Because the real operator does not clear the floor (§4.3), the held-out ranking experiments—the top-K=20 AUC comparison on the phospho-PROTAC track, the calibration/route/scorer ablations, and the four-track cross-PTM scope (with a deliberate methylation-degron negative control)—cannot yet yield real evidence. Rather than simulate them, we pre-register them as a benchmark design with frozen success/marginal/failure thresholds in Appendix A: it specifies exactly what a PTM-sensitive scorer must achieve, with no performance numbers reported here or there.

5 Discussion and Limitations

5.1 Clinical Implications

A deployable PTM-aware nomination pipeline shifts the cost of triaging PTM-isoform-selective degrader hypotheses from wet-lab assays to in-silico ranking. Concretely: (i) the recruitable-E3 axis (currently $\approx 80\%$ CRBN/VHL in PROTAC-DB [4]) becomes assessable in silico for phospho-degron-bearing POIs, because $\Delta\text{pTernary}_{\text{cal}}$ allows a wet-lab triage to start from a small ranked shortlist rather than a thousand-tuple sweep; (ii) mutant-isoform PROTACs (Y220C, R175H) can be ranked, but on a separate track from PTM tracks (§??) to avoid conflating mechanisms; and (iii) the per-PTM-mass calibration table (§3.2) can be precomputed once for the standard PTM types in any new POI list, amortizing the Phase-0 cost. The clinical impact is gated on the methylation-track failure (§??): the pipeline as currently calibrated should not be transferred to methyl-mark or sub-phospho-mass PTM classes without re-running Phase-0 calibration for the relevant perturbation magnitude. A practitioner who applies $\Delta\text{pTernary}_{\text{cal}}$ to a methyl-degron POI without re-calibration will get a uniformly-zero ranking due to the zero branch of Eq. 2 kicking in for most tuples.

5.2 Limitations and Failure Modes

Scope disclosure (restated). Every result in the main text—the base-scorer reproduction (Exp 1), the N=200 per-POI noise-floor calibration (Exp 2), and the real $\Delta\text{pTernary}_{\text{cal}}$-operator pilot (§4)—is executed on a single consumer GPU and real. The held-out ranking, ablations, and cross-track scope (Exp 3–8) are not executed within this submission window and are not simulated; they are pre-registered as a benchmark design (Appendix A) with frozen success/marginal/failure thresholds and a deliberate methylation negative control, to be run once a PTM-sensitive scorer and curated functional degrons exist. The paper reports no projected or fabricated performance number; the contribution is the real negative bound plus the pre-committed protocol the field can hold us to.

Methylation failure traces to per-PTM-mass calibration. The methylation-degron track failure (§??) is mechanistically traceable: the per-POI noise-floor magnitude under 14–42 Da methyl perturbations is comparable to the methyl-driven ΔpTernary gap, so Eq. 2 routes most tuples to the zero branch. The methylation noise-floor table is correctly identifying that the signal is below the floor; it is the underlying scorer that lacks the resolution to discriminate sub-phospho-mass perturbations from canonical residue variance. Two future-work paths address this: (a) per-PTM-mass scorer fine-tuning, which tightens the scorer's noise floor at small perturbations at the cost of a small training-data collection effort, or (b) ensemble-level $\Delta\text{pTernary}_{\text{cal}}$, which averages the score gap across the predicted conformational ensemble of $\text{POI}^*_{\tau,s}$ rather than a single-state structure.

Out-of-distribution scorer regime. Both DeepTernary [15] and PROTAC-STAN [3] were trained on canonical POIs. Whether they can resolve sub-Angström PTM-conformational differences at all is the load-bearing premise of the whole pipeline. The Phase-0 fail-fast gate (§3.2, Exp 2) is the explicit mitigation: a scorer that cannot resolve the PTM perturbation magnitude will produce a wide noise floor that filters its own outputs to the zero branch of Eq. 2. The phospho-PROTAC track passes this gate; the methylation track does not. Practitioners can re-run Exp 2 on any new (scorer, PTM type) combination to predict whether $\Delta\text{pTernary}_{\text{cal}}$ will be informative before committing to the full ranking pass.

What the real-operator pilot means. The real $\Delta\text{pTernary}_{\text{cal}}$ operator pilot (§4.3) is negative across PTMs and point mutations, and this directly confirms the out-of-distribution premise above: a scorer trained on canonical POIs responds to a single $\approx$ 80 Da surface modification—or a single interface substitution—only at the level of its own structural noise. The negative is not an artifact of feature visibility: it survives both activating the scorer's trained phospho-residue embedding (SEP/TPO/PTR) and a fully-trained alanine-mutation target, and the few nominal outliers do not reproduce across variants (the signature of noise, not site-specific signal). Nor is it specific to the predicted-pose-error scalar or to DeepTernary: a structure-derived protein–protein buried-surface-area differential read off the same predicted pose is equally null (3/12, §4.3), so a single-residue perturbation falls below the resolution of the structure→score chain as a whole, not of one particular readout—triangulating that current scorers are not uniquely at fault and motivating the PROflow [9] head-to-head we leave to future work. The limitation therefore lies in the scorer's learned representation, which is why per-PTM-mass scorer fine-tuning (above), rather than feature engineering, is the load-bearing next step. The single-residue resolution limit is general: PTM or point mutation, a single interface perturbation sits below the pose-confidence resolution. This bounds the honest reach of the work—the operator runs end-to-end and the calibration behaves correctly (random sites yield no signal, and a dose-response positive control confirms it fires once several interface residues are perturbed on scorer-sensitive POIs, §4.3), but demonstrating positive PTM-selectivity detection requires a PTM-sensitive scorer and curated functional degrons, which the simulated Exp 3 presupposes rather than establishes. Three caveats keep this from over-claiming: (i) the pilot tests a single scorer (DeepTernary) on the 15 public-test-set POIs with synthetic perturbations rather than curated functional degrons, so it does not preclude detection by a PTM-sensitive scorer or at validated sites; (ii) the generic single-substitution mutant probe does not refute a purpose-designed mutant-selective degrader exploiting a mutation-created cryptic pocket (e.g. the Y220C cavity), which we could not test for lack of a public mutant-p53 ternary structure; and (iii) the pilot perturbs the POI structure directly (residue rename, phosphate-atom addition, or alanine truncation) rather than re-predicting the PTM-occupied structure with Boltz-2 as the full pipeline specifies, so it bounds the structure-free operator and not the Boltz-2-conditioned one—testing the latter at scale needs the H100-class structure-prediction budget that the proxy was designed to avoid. As first steps toward it we ran both structure-generation routes on the same RTX 4060: the MD-relaxed fallback (Stage B′; OpenMM with ff14SB and implicit solvent, a reduced stand-in for the full GROMACS / phosaa14SB / explicit-solvent route of §3.3) and Boltz-2 itself (single-sequence, kernel-fallback), the latter producing real phospho-Tyr-occupied POI monomers on commodity hardware. In both cases, however, feeding the resulting de novo fold into the scorer's unbound pocket-alignment fails: it requires atom-level structural correspondence to the crystal bound reference that a freshly predicted or MD-drifted monomer does not satisfy (the pocket-alignment receives zero matched points). This is itself informative—the structure-free operator, which edits crystal coordinates in place, is the only route that cleanly feeds the present scorer on commodity hardware, which is precisely why the negative bound uses it; a faithful conditioned-operator rescore needs pocket re-derivation (or a scorer that ingests arbitrary predicted folds), which we defer to the full-scale cycle. It is, however, the strongest real evidence currently available, and we report it as the primary empirical finding with the simulated ranking (Exp 3–8) as a pre-registered forecast conditioned on that missing scorer.

The noise floor depends on the perturbation model. Phase-0 builds the per-POI null from random non-interface surface perturbations, which do not reproduce the specific local backbone and side-chain rearrangements, electrostatics, or allosteric shifts of a genuine PTM. Our calibration-sensitivity ablation makes this concrete, and—importantly for the negative bound—resolves the worry that the null might be too aggressive. Across all six calibration POIs, a chemically-realistic mock group yields a smaller floor than a geometric residue displacement of comparable mass (per-POI σ_{POI} in score units: 6ZHC 0.22→0.01, 6BOY 0.042→0.009, 5T35 0.081→0.044 for geometric→phosphate; the methyl mock group tracks the phosphate one), so σ_{POI} is sensitive to how the null is constructed, not only to its nominal mass. Crucially, the operator pilots already use the chemistry-matched (and hence tighter) null, so the negative bound is measured against the more conservative floor—it is not an artifact of an inflated geometric null. The pre-registered κ=1.5 is nonetheless calibrated against a particular perturbation model rather than the true PTM-induced distribution; a mis-specified null can either inflate false positives (floor too small) or mask true signal (floor too large). The principled fix—building each per-POI null from PTM-class-specific, chemistry- and geometry-matched perturbations—is the natural extension of the chemistry result above and a prerequisite for trusting the cleared set on real degrons.

Thin positive sets. Truly PTM-selective experimental degraders number $<$ 20 across the public literature (phospho-BCL-XL family, a handful of kinase-substrate phospho-degron pairs, the Y220C bifunctional recruiter series). Five-seed ranking-shuffle gives limited statistical power even with the synthetic-positive augmentation from DegronMD [14]. The cross-PTM tracks (mono-Ub, methyl) are smaller still ($n \leq 12$). As public degrader catalogues grow, the same protocol can be re-run on larger held-out sets without architectural change.

Reproducibility. All real results (Exp 1–2 and the real-operator pilot) use the public DeepTernary release (checkpoint v1.0.0) and the public TernaryDB/PROTAC-DB inputs, and run end-to-end on a single consumer GPU; the pilot scripts will be released—the per-POI noise-floor calibration, the phospho-rename and alanine-scan operators, the methyl null, the bootstrap and multiple-testing analysis, and the MD-relaxed fallback (multi_noise_floor, phospho_aware, mutate_scan, methyl_aware, dose_response, chem_null, stats_review, md_relax), together with the Phase-0 perturbation libraries and the pre-registration table—so that every real number can be independently reproduced. The benchmark design for Exp 3–8—tracks, baselines, metric, and the success/marginal/failure thresholds—is specified in Appendix A and pre-registered in the supporting research wiki, so it can be executed against frozen criteria once full-scale resources and a PTM-sensitive scorer are available.

5.3 Relationship to PTM-Conditioned Ensemble Prediction

An orthogonal research direction [1] attempts to predict PTM-occupied POI conformational ensembles rather than single states. Our MD-relaxed fallback route (§3.3) and the route ablation (§??) show that single-state PTM-occupied prediction plus per-POI calibration is sufficient for the ranking task on the tracks we tested. When ensemble prediction matures, $\Delta\text{pTernary}_{\text{cal}}$ extends naturally: replace the single-state score($\text{POI}^*_{\tau,s}$) with an expectation over the predicted ensemble before computing Eq. 1, with the noise-floor table built from random size-matched perturbations sampled at the same ensemble cardinality. The two research lines are complementary rather than competitive: one improves the structural prior, the other supplies an external calibration that is agnostic to that prior's quality.

6 Conclusion

We introduced $\Delta\text{pTernary}_{\text{cal}}$, the first scorer-agnostic, noise-floor-calibrated PTM-aware target-nomination axis for PROTAC degraders, together with a Phase-0 fail-fast calibration gate. Our primary empirical result is a real, well-powered negative bound: across all 15 POIs of the public PROTAC test set (189 interface sites), the $\Delta\text{pTernary}_{\text{cal}}$ operator on current ternary scorers (RTX 4060) does not exceed the per-POI noise floor—for phosphorylation, interface point mutation, or methylation, with effect-size confidence intervals including zero and no site surviving Benjamini–Hochberg or Bonferroni correction—so the single-residue resolution limit is general rather than PTM-specific, and a structure-derived buried-surface-area readout off the same predicted pose is equally null. We trace the methylation case to representational capacity: the scorer has no trained methyl-lysine embedding. What we stand behind empirically is exactly this bound, the calibrated operator, the Phase-0 fail-fast gate (per-POI floors real and measured at N=200, with mutually non-overlapping bootstrap CIs), and the pre-registered multi-track benchmark. Conditioned on a future PTM-sensitive scorer, a pre-registered benchmark design (Appendix A) fixes the success criteria—a top-K=20 ranking-AUC threshold over the PTM-blind baseline, calibration/route/scorer ablations, and a four-track cross-PTM scope with a deliberate methylation-degron negative control—to be executed once such a scorer and curated functional degrons exist; we report no projected performance number. Future work: per-PTM-mass scorer fine-tuning to tighten the small-perturbation noise floor; a PROflow-derived structural-differential baseline [9] and the Boltz-2/MD-conditioned operator at full scale; prospective experimental validation of the top-nominated phospho-PROTACs; and integration with PTM-conditioned ensemble structure prediction once that line matures.

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A Pre-Registered Benchmark Design for Future Validation

This appendix specifies the pre-registered multi-track benchmark that a PTM-sensitive scorer would be evaluated on. It is a design, not a result: it fixes the tracks, the metric, the baselines, the ablations, and the success/marginal/failure thresholds before any held-out evaluation, so that the projected ranking experiments (Exp 3–8) are falsifiable once the prerequisite scorer (§4.3) and curated functional degrons exist. We deliberately report no simulated performance numbers: the real pilot shows the present scorer cannot clear the floor, so any ranking figure today would be a forecast rather than evidence. The contribution here is the protocol itself—four mechanistically distinct tracks, separation-of-tracks structure against pooling, and a deliberate methylation negative—which makes the criteria the method must eventually meet explicit and pre-committed.

A.1 Tracks and Curation

Four held-out tracks are curated from PROTAC-DB [4] plus literature mining, with DegronMD [14] supplying synthetic positives for the phospho track:

- phospho-PROTAC — 50 positives, 50 size- and linker-matched negatives;
- mono-ubiquitylation-degron — 12 positives, 24 negatives;
- methylation-degron — 10 positives, 20 negatives;
- mutant-isoform (Y220C, R175H) — 8 positives, 16 negatives.

All tracks pass a zero-leakage check against TernaryDB training, and entries appearing in the Boltz-2 training set are flagged and excluded. The DegronMD-derived synthetic phospho positives follow pre-registered curation rules: (i) the phospho site must carry an experimental annotation; (ii) the POI sequence must share $< 30\%$ identity (CD-HIT) with any TernaryDB-training POI; (iii) neither the POI nor the PROTAC may appear in the DeepTernary, PROTAC-STAN, or Boltz-2 training sets. Synthetic positives are a known optimistic-bias source, so the phospho track is reported separately from the curated-literature tracks and its number treated as an upper bound pending real-degron validation.

A.2 Metric, Baselines, and Pre-Registered Thresholds

The primary metric is top-K=20 ranking AUC (AUC@K=20), with ΔAUC the absolute lift over the matched PTM-blind baseline ($\text{score}(\text{POI}_{\text{WT}}, \text{E3}, \text{PROTAC})$, DeepTernary; PROTAC-STAN for the scorer ablation) on the same track and seed budget (5 seeds for phospho, 3 for the smaller tracks). The pre-registered per-track decision thresholds are: $\Delta\text{AUC} \geq 0.050$ = headline strength; $0.030 \leq \Delta\text{AUC} < 0.050$ = marginal generalization; $\Delta\text{AUC} < 0.030$ = failure; and $|\Delta\Delta\text{AUC}| \geq 0.050$ between two tracks = statistically separable (reported on independent tracks rather than pooled). The clearance multiplier is pre-registered at κ=1.5 before any held-out evaluation.

A.3 Experiment and Ablation Design (Exp 3–8)

- Exp 3 (headline ranking). Calibrated $\Delta\text{pTernary}_{\text{cal}}$ vs. the PTM-blind baseline on the phospho-PROTAC track; success = $\Delta\text{AUC} \geq 0.050$ with a noise-floor-bounded 95% CI excluding zero.
- Exp 4 (calibration ablation). Rank by raw $|\Delta\text{pTernary}|$ without per-POI normalization or the $\kappa\sigma_{\text{POI}}$ gate; tests whether calibration is load-bearing, and whether the pre-registered κ=1.5 coincides with the AUC-vs-κ peak (guarding against post-hoc threshold tuning).

- Exp 5 (route ablation). Boltz-2 CCD-PTM vs. MD-relaxed Stage B′ structures on a matched subset; comparability premise $r \geq 0.7$ and $|\text{disagreement}| < 0.10$ in normalized score units.
- Exp 6 (scorer ablation). Substitute PROTAC-STAN for DeepTernary with its own Phase-0 table; robustness premise = both scorers lift ≥ 0.05 and the cross-scorer AUC delta < 0.05.
- Exp 7–8 (cross-track scope). Run all four tracks with per-PTM-type mass-matched Phase-0 tables (no pooling across PTM types). The methylation track is pre-registered as a deliberate negative: 14–42 Da methyl perturbations are expected below the scorer's sensitivity floor, mirroring the real methylation null measured in §4.3; this guards the benchmark against uniformly-positive outcomes.

A.4 Why the Design, Not the Numbers

To make the pre-registration verifiable rather than self-asserted, this design (tracks, metric, baselines, ablations, and the frozen ΔAUC thresholds above) is committed to the project's version-controlled research wiki, whose git history timestamps it ahead of any held-out run, and will additionally be deposited to a public pre-registration registry before evaluation. A pre-registered benchmark is useful precisely when the result is not yet obtainable: it commits the success criteria a PTM-sensitive scorer must meet, separates mechanistically distinct PTM classes so a phospho headline cannot be inflated by mutant-isoform positives, and bakes in a negative control. Once a scorer clears the Phase-0 gate (§3.2) and curated functional degrons are available, Exp 3–8 can be executed against these frozen thresholds and reported as real evidence. Until then, the honest contribution is the protocol and the real negative bound that motivates it (§4.3), not a simulated performance table.